

Eric R. Xu, Ph.D.

SENIOR DATA & AUTOMATION ENGINEER | PYTHON · VALIDATION FRAMEWORKS · WORKFLOW AUTOMATION

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PROFESSIONAL SUMMARY

Senior data and automation engineer and technical lead with 10+ years building Python automation, validation frameworks, and production data services at Apple, Meta, DoorDash, and Inmar. Hands-on with CI/CD (Git, Buck), pytest testing, and containerized deployment (Docker, Kubernetes). Managed two direct-report analysts, mentored new hires, and set and tracked team metrics with data scientists using A/B tests with holdouts. Owns solutions end to end and presents technical direction to stakeholders and management, including in marketing operations.

TECHNICAL SKILLS

Automation & Engineering: Python · pytest (unit & integration) · API integration · workflow orchestration (Airflow) · webhook-driven integrations · Slack Bolt / Block Kit · diff review

CI/CD & Infrastructure: Git · Buck · PR validation & deployment · Docker · Kubernetes (CronJobs, service deployments) · DigitalOcean · AWS S3

Data & Platforms: PostgreSQL · MySQL · Snowflake · Elasticsearch · Spark · Hive · Presto/Trino · Vector Search (Qdrant) · dimensional modeling

Quality, Metrics & AI: monitoring/alerting · anomaly detection · data-quality validation · A/B testing · Metabase · Tableau · Looker · LLM/RAG workflows

EXPERIENCE

TileSense — Mahjong Trainer (Independent Project)

2026 – Present

- Designed a mahjong-strategy model that extends and aggregates existing tile-efficiency and expected-value methodologies to more accurately weigh all factors relevant to a player's decision, empirically outperforming existing OpenRiichi-project bots.
- Built and shipped a cross-platform Flutter app with a PostgreSQL backend and a Docker-containerized multiplayer service on DigitalOcean, instrumented with in-app eventing to capture gameplay signals, and built Metabase dashboards over the app's PostgreSQL event tables tracking Daily Active Devices and retention.

Apple — Data Engineer (contracted by TEKsystems)

03/2025 – 09/2026

- Built and supported Python/SQL (Presto/Trino) CronJob pipeline containers on Kubernetes — including webhook-driven Slack (Bolt) alert-enrichment and RAG-based triage services — improving real-issue detection and coverage by 35% YoY (false positives down 30–40%) and increasing alert engagement by 20% at stable volume.
- Partnered with Engineering, Product, and marketing stakeholders to translate ambiguous requirements into metric definitions, scalable pipelines, Tableau reporting, and clear technical recommendations.

DoorDash — Analytics Data Engineer (contracted by TEKsystems)

08/2024 – 02/2025

- Designed layered analytics transformations and serving datasets in Snowflake, and self-service BI for in-store deals, supporting market expansion, redemption, conversion, and cohort analysis.
- Defined reusable metric logic, automated data-quality checks, and experiment launch monitoring, informing UI strategies that improved conversion efficiency by 5–10%; built forecasting and milestone tracking across 50K+ redemptions, reducing ad-hoc analysis.

Meta — Senior Analytics Data Engineer

07/2020 – 04/2024

- Designed and owned Spark/Hive/Presto data pipelines and dimensional (star-schema) tables in a layered raw → curated → serving architecture, supporting analytics, experimentation, and high-traffic consumer products.
- Shipped changes through Git/Buck PR review and deployment, backed by pytest unit and integration tests and diff-review practices; mentored new hires and partnered with Meta's Safety (Integrity) team and London-based colleagues across time zones on impersonation-detection initiatives.
- Built reusable reporting datasets and KPI frameworks — including a CPU-efficiency metric that drove a 15% increase in efficiency-focused code changes and reduced compute costs (before/after analysis) — and partnered with Data Scientists to set team-level metrics and measure feature impact via A/B tests with test and control holdouts.

Inmar Intelligence I OwnerIQ — Senior Data Engineer / Senior Data Scientist

07/2015 – 06/2020

- Managed a data product analytics team (2 direct reports) for 3 years, leading end-to-end delivery of 8+ production data products with stakeholder acceptance.
- Built large-scale transformation, recommendation, and data products batch pipelines with Python, Airflow, Spark, Hive, Presto, AWS S3, MySQL, and Elasticsearch k-NN to generate commerce insights and ROAS gains.
- Implemented automated data-quality checks, Looker KPI reporting, and internal analytics tools that standardized commerce measurement and reduced manual analysis.

EDUCATION

Certificate, Data Science & Machine Learning — MIT Schwarzman College of Computing, Nov 2024

Ph.D., Psychological & Brain Sciences — Dartmouth College

B.S., Biomedical Engineering (Biomechanics) — Case Western Reserve University